INCH-POUND

MIL-T-3351E
22 February 1990
SUPERSEDING
MIL-T-3351D
10 October 1975

MILITARY SPECIFICATION

TRACTOR, FULL-TRACKED, LOW-SPEED; TRACTOR WHEELED, AGRICULTURAL; AND TRACTOR WHEELED, INDUSTRIAL; AND THEIR ATTACHMENTS, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the packaging (preservation, packing, and marking) of tractors, full-tracked, low-speed, wheeled agriculture and wheeled industrial and their attachments.

1.2 Classification. Preservation, and packing shall be one of the following type, as specified (see 6.2):

Type I - Boxed.
Type II - Unboxed (mobile).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A AREA PACK
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer Coating, Alkyd, Corrosion-inhibiting, Lead and Chromate Free, Voc-compliant.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-L-800	- Lubricating Oil, General Purpose, Preservative, Water Displacing, Low Temperature.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-S-196	- Support Items, Accessories and Kits, Mechanical: Packaging of.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-C-5501	- Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-E-16298	- Electric Machine Having Rotating Parts, Accessories and Associated Support Items: Packaging of.
MIL-P-52905	- Paint, Arctic, Camouflage, Removable.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC., AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein and in applicable specifications. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects.

3.2 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The first article pack may be accomplished utilizing either the first article tractor or a production tractor. If the first article tractor is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article tractor and production tractors.

3.3 Repainting. Painted surfaces of the tractors and attachments on which the paint has been damaged shall be cleaned and repainted in accordance with the painting requirements of the end item specification (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

3.4 Preservation. Preservation shall be level A or C, as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 Disassembly. Disassembly shall be confined to the minimum necessary to safeguard parts vulnerable to damage, pilferage and loss, except as required by the transportability requirements of the end item specification. The extent of further disassembly shall be confined to the removal of protruding parts that would otherwise increase cubage; however, disassembly shall be limited to those parts that can be removed and reinstalled without special tools. Removed bolts, nuts, screws, pins, and washers shall be reinstalled in mating parts and secured to prevent their loss. When it is not practical to reinstall the hardware, the items shall be preserved in accordance with MIL-P-116, method IC-1, and the preserved items secured to the component from which they had been removed. Keys removed shall be secured in the keyway of the primary component or shaft.

3.4.1.2 Matchmarking. Parts removed and mating parts on the tractors and attachments shall be matchmarked when necessary to facilitate reassembly. Large parts shall be matchmarked identically by stenciled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121, or identically matchmarked with paint conforming to MIL-P-52905. Small parts removed and mating parts on the tractors shall be matchmarked and identified with tags as specified in 3.4.1.3.

3.4.1.3 Tags. Tags used for matchmarking and instructions shall conform to UU-T-81, type A, or type B, class 2. Marking of the tags shall be with waterproof ink.

3.4.1.4 Cleaning, drying, and preservatives. Metal surfaces to which a preservative is to be applied shall be cleaned and dried in accordance with MIL-P-116. Preservatives identified herein by "P" numbers shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

3.4.1.5 Unprotected metal surfaces. Unpainted exterior metal surfaces of tractors and attachments, except machined surfaces, requiring the application of a contact preservative in accordance with MIL-P-116, such as tie rods, adjusting rods, clevises, bolts, nuts, coil springs, leaf springs, cable sheaves pins, exposed linkages, grease fittings, drawbars, and any other surface including those surfaces exposed by disassembly, shall be coated with type P-1 preservative. Exterior machined surfaces, where removal is required to place the tractor in operation, shall be coated with type P-6 or P-11 preservative. Other machined surfaces, where removal is not required, shall be coated with type P-1 preservative. When applicable, surfaces coated with type P-6 or P-11 preservative shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured in place with tape conforming to PPP-T-60, type IV, class 1.

3.4.1.6 Engines. Engines and engine accessories shall be preserved in accordance with MIL-E-10062, level A, type II, method I, for boxed tractors, and MIL-E-10062, level A, type I, method I, for unboxed (mobile) tractors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

3.4.1.7 Winterization components.

3.4.1.7.1 Pumps, burners, torches, and piping. Interior surfaces of pumps, burners, torches, and piping requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-10, type I, grade 10 preservative and the excess preservative allowed to drain. The threaded surfaces of fillcaps and drainplugs shall be coated with the preservative specified herein and the caps and plugs reinstalled. Openings in burners not closed with caps and plugs shall be closed with tape conforming to PPP-T-60, type IV, class I. Blowtorches and blowtorch-type burners shall each be placed in a fiberboard box conforming to PPP-B-636, W5c or W6c. The box shall be closed in accordance with method V of the appendix to the box specification.

3.4.1.7.2 Fuel tanks. The fuel tanks for heater fuel, if not equipped with baffles, shall be fogged with type P-10, type I, grade 10 preservative, and the excess preservative drained. Fuel tanks with baffles shall be completely filled with the preservative and drained. The drainplug and fillcap threads shall be coated with the preservative, and the drainplug and fillcap shall be reinstalled. The preservative may be reused for processing other tanks but shall be discarded when more than 10 percent of the resultant mixture is fuel (see 6.5).

3.4.1.7.3 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298, level A, using the alternate method required for equipment assembled to machines.

3.4.1.8 Drive shafts and universal joints. Unprotected surfaces of drive shafts and universal joints shall be coated with type P-1 preservative. Dust boots shall be reinstalled after the preservative has dried.

3.4.1.9 Gears.

3.4.1.9.1 Exposed gears. Unprotected surfaces of exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

3.4.1.9.2 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with preservative conforming to type P-10, type I, grade as applicable. The gear housing shall be tagged to indicate: "The preservative oil contained in this gear housing is good for operation. Do not drain until the first required lubricant change." Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with the applicable grade of lubricant conforming to MIL-L-2105. The gear housing shall be tagged to indicate: "The lubricant contained in this gear housing is good for operation. Do not drain until the first required lubricant change." Gears not operating on lubricating oil (SAE 10, 30, or 50) or gear lubricant (SAE 80 or 90) shall have the housing filled to the operating level with the approved lubricant required for operation. The gear housing shall be tagged to indicate: "This gear housing is filled to the operating level with the applicable type lubricant required for operation. Do not drain until the first required lubricant change." Tags shall be as specified 3.4.1.3.

3.4.1.10 Drive chains.

3.4.1.10.1 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to insure penetration of the preservative to

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

the inner surfaces of the rollers, pins and bushings. The entire chain shall be coated with type P-1 preservative after the excess preservative has drained.

3.4.1.10.2 Enclosed drive chains. Enclosed drive chains shall be preserved as specified in 3.4.1.9.2 for enclosed gears.

3.4.1.11 Clutches. Clutches shall not be operated after being preserved as specified herein.

3.4.1.11.1 Disc type (dry type). With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. When specified (see 6.2), the clutch shall be disassembled sufficiently to permit the application surfaces of a film of the same primer to friction surfaces of the pressure plates. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall have the clutch pedal secured in a partially disengaged position to eliminate contact between the disc facings and the pressure plates. Snap-over-center- and toggle-in-type clutches shall be completely disengaged. Cover plates shall be reinstalled.

3.4.1.11.2 Band and shoe type. Clutch drum facings and control mechanisms enclosed in the housing shall be coated with primer conforming to TT-P-664. Clutch control mechanisms not enclosed in housing shall be coated with type P-1 preservative.

3.4.1.11.3 Cone type. The unlined cones and control mechanisms enclosed within housings shall be coated with primer conforming to TT-P-664. Control mechanisms not enclosed in housings shall be coated with type P-1 preservative. The clutch shall be disengaged to prevent the linings from adhering to the unlined cones.

3.4.1.11.4 Jaw type. Clutch jaws, shifter yokes, and machined surfaces of shafts shall be coated with type P-1 preservative. Control mechanisms enclosed within housings shall be coated with type P-10, type I, grade 10 preservative. Control mechanisms not enclosed within housings shall be coated with type P-1 preservative.

3.4.1.11.5 Flywheel clutch, oil type (running in oil). The clutch shall be preserved at the same time as the engine preservative run-in. The housing shall be filled to the operating level with preservative lubricating oil conforming to type P-10, type I, grade as applicable. The fill plug threads shall be coated with the preservative lubricating oil and the fill plug reinstalled. When the engine reaches its normal operating temperature, the clutch shall be operated to insure coating of all parts and surfaces with the preservative. Upon completion of preservation, all openings into the clutch housing shall be sealed with tape conforming to PPP-T-60, type IV, class 1. A tag shall be attached to the clutch control indicating: "This clutch housing is filled to the operating level with preservative lubricating oil, applicable grade, and is suitable for operation until the first required lubricant change. Remove tape from openings prior to operation of the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.6 Flywheel clutch, oil type (lubricated by engine lubricating system). The clutch shall be processed at the same time as the engine preservation run-in. When the engine reaches its normal operating temperature, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

clutch shall be operated to insure coating of all parts and surfaces with the lubricant. Upon completion of processing, all openings into the clutch housing shall be sealed with tape conforming to PPP-T-60, type IV, class 1. A tag shall be attached to the clutch control indicating "Remove tape from openings prior to operation of the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.7 Steering clutches.

3.4.1.11.7.1 Dry type. Dry disc-type steering clutches shall be preserved as specified in 3.4.1.11.1. Other dry-type steering clutches shall be preserved as specified herein for the applicable type of dry clutch.

3.4.1.11.7.2 Oil type. The clutch housing shall be filled to the operating level with preservative lubricating oil conforming to type P-10,type I, grade as applicable. The fill plug threads shall be coated with the preservative lubricating oil and the fill plug shall be reinstalled. The clutch shall be operated to insure coating all parts and surfaces with the preservative. Upon completion of preservation, all openings into the clutch housing shall be sealed with tape conforming to PPP-T-60, type IV, class 1. A tag shall be attached to the clutch control indicating: "The housing is filled to the operating level with preservative lubricating oil of applicable grade, and is suitable for operation until the first required lubricant change. Remove tape from openings prior to operation of the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.7.3 Hydraulic boosters (having a separate oil supply). Hydraulic boosters having a separate oil supply reservoir shall be preserved at the same time as the steering clutch. The oil reservoir shall be filled to the operating level with preservative lubricating oil conforming to type P-10, type I, grade as applicable. The fill plug threads shall be coated with the preservative lubricating oil and the fill plug reinstalled. The booster shall be operated to insure coating all parts and surfaces with the preservative. Upon completion of preservation, all openings into the system shall be sealed with tape conforming to PPP-T-60, type IV, class 1. A tag shall be attached to the clutch control indicating: "The hydraulic booster is filled to the operating level with the applicable grade of preservative lubricating oil and is suitable for operation until the first required lubricant change. Remove tape from openings into the system prior to operating the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.8 Clutch housing. For type I boxed tractors, drainplugs for dry-type clutches shall be removed from clutch housings. The plugs shall be identified by tags specified in 3.4.1.3. The threads of the plugs and the housing threads shall be coated with preservative lubricating oil conforming to type P-10, type I, grade 10. The plugs shall be secured in a protected and conspicuous location. A tag specified herein shall be secured to the clutch control indicating: "Reinstall drainplug in clutch housing prior to operation of the clutch." For type II unboxed (mobile) tractors, openings into clutch housing shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured in place with tape conforming to PPP-T-60, type IV, class 1. A tag as specified herein shall be attached to the clutch control indicating: "Remove covers from housing openings prior to operation of the clutch."

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

3.4.1.12 Brakes.

3.4.1.12.1 General. Exterior, unpainted, untreated, and threaded surfaces of cables, clevises and linkages of service and parking brakes requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. When specified (see 6.2), metal surfaces enclosed within brakedrums (such as adjusting wedges, pins, eccentrics, cam levers, linkages, anchors, retracting springs) and the breaking surface of the brakedrums shall be sprayed with a continuous film of primer conforming to TT-P-664. The film shall be a minimum of 0.50 mil to a maximum of 0.75 mil thick as determined by an appropriate film thickness gauge. Primer material containing lumps or skin that will not return to a homogenous mixture upon stirring shall not be used. The film shall be thoroughly dry prior to reinstalling the brakedrums. A warning tag shall be securely attached to the brake control indicating: "Brakedrums and components preserved. Do not apply brakes when tractor is being moved. Towing shall be accomplished by use of a rigid towbar or similar arrangement until this tractor is issued to the user." Tags shall be as specified in 3.4.1.3.

3.4.1.12.2 Hydraulic brake system.

3.4.1.12.2.1 Systems operating on hydraulic brake fluid. The hydraulic brake system shall be filled to the operating level with brake fluid specified in the vehicle specification and the brakes actuated to insure coating all interior parts and surfaces with the brake fluid.

3.4.1.12.2.2 Systems operating on petroleum base fluid. The hydraulic systems of brakes and steering systems that operate on petroleum base hydraulic fluids shall be filled to the operating level with the fluid required for operation and the brakes and steering system shall be actuated to insure coating of all interior surfaces with the fluid. A tag shall be attached to the system controls indicating: "This system is filled with the approved operating fluid. Do not drain." Tags shall be as specified in 3.4.1.3.

3.4.1.12.3 Air-actuated brake systems. The air-actuated brake systems shall be drained. Interior surfaces of the air-supply tanks shall be fogged with type P-10, type I, grade 10 preservative. Any excess preservative lubricating oil shall be drained. Threaded openings and the threads of drainplugs shall be coated with preservative lubricating oil specified herein. The drainplugs shall be reinstalled. Air-line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust valves shall be closed by inserting plugs conforming to MIL-C-5501 or by sealing the ports with tape conforming to PPP-T-60, type IV, class 1. Tags shall be attached to the valves and service lines indicating: "The plugs or tape must be removed from the exhaust ports and valves before this tractor is placed in operation." Tags shall be as specified in 3.4.1.3.

3.4.1.12.4 Air-hydraulic brakes. Air-hydraulic brakes shall be preserved in accordance with 3.4.1.12.1, 3.4.1.12.2, 3.4.1.12.3. To preserve the interior surfaces, the plugs shall be removed from housing and connectors and the preservative lubricating oil shall be sprayed through each opening. The plugs and hose shall be reinstalled, and the brake pedal shall be operated through two

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

cycles. The plugs and hose shall again be removed, and the interior surfaces shall again be sprayed as specified herein. The plugs and hose shall again be reinstalled.

3.4.1.13 Hydraulic systems (except hydraulic brakes or steering systems). The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches on the ram cylinders. The wraps shall be secured in place with tape conforming to PPP-T-60, type IV, class 1. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. Any unpainted exterior metal surfaces of the system requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. A tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Tags shall be as specified in 3.4.1.3.

3.4.1.14 Automatic transmissions, fluid couplings, and torque conversion transmissions.

3.4.1.14.1 Automatic transmissions and fluid couplings. The housings shall be filled to the operating level with fluid or lubricant recommended for operation. The transmissions and fluid couplings shall be operated at sufficient engine speed to insure coating of all interior surfaces with the fluid or lubricant. Openings into the housing shall be sealed with tape conforming to PPP-T-60, type IV, class 1. A tag shall be attached to the control lever indicating: "The housing is filled to the operating level with the approved fluid or lubricant and is good for operation of this tractor. Do not drain. Do not operate tractor until the tape has been removed from the openings." Tags shall be as specified in 3.4.1.3.

3.4.1.14.2 Torque conversion transmissions.

3.4.1.14.2.1 Lubricating oil operated. The converter shall be preserved at the same time as the engine. At the beginning of engine preservation, the converter housing shall be filled to the operating level with the approved operating oil. The converter shall be operated a sufficient length of time during fuel system preservation to insure coating of all interior surfaces with the operating oil. Openings into the system shall be sealed with tape conforming to PPP-T-60, type IV, class 1. A tag shall be attached to the converter control indicating: "The converter housing is filled to the operating level with the approved oil required for operation. Do not drain. Remove tape from housing openings prior to operation of this tractor." Tags shall be as specified in 3.4.1.3.

3.4.1.14.2.2 Diesel fuel operated. The converter shall be preserved at the same time as the engine. At the beginning of engine preservation, the system shall be drained of diesel fuel. The converter system shall be filed to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

operating level with type P-9 preservative and the converter operated a sufficient length of time during fuel system preservation to insure coating of all interior parts with type P-9 preservative. The type P-9 preservative shall remain in the system. Openings into the system shall be sealed with tape conforming to PPP-T-60, type IV, class 1. A tag shall be attached to the converter control indicating: "The converter system is filled to the operating level with type P-9 preservative. Do not operate this tractor until the type P-9 has been drained and the system has been filled to the operating level with the approved operating fuel. Remove tape from openings into housing." Tags shall be as specified in 3.4.1.3.

3.4.1.15 Wheels, axles and bearings. Wheels and hubs, if removed from the axles or spindles for type I boxed tractors, shall have the interior bearing surfaces of the axles and spindles coated with type P-6 or P-11 preservative. Coated surfaces of the axles or spindles shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wrap or cover shall be secured in place with tape conforming to PPP-T-60, type IV, class 1. Bearings removed shall be coated with type P-6 or P-11 preservative and shall be individually preserved in accordance with MIL-P-116, method IC-1. Openings of the hubs shall be covered with barrier material as specified herein and the barrier material secured in place with the tape specified herein.

3.4.1.16 Rubber tires. Tires mounted on the wheels and rims (road tires) shall be inflated to maximum operating pressure for maximum load. Tires mounted on wheels and rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

3.4.1.17 Crawlers. Surfaces of unpainted metal crawler treads, tracks, sprockets, bogie wheels, rolls, pins, linkages, grouser tips, roller surfaces, and link rails that are readily accessible shall be coated with type P-1 preservative. However, absence of preservative on those components will not impair their usefulness.

3.4.1.18 Instrument panels. For type II unboxed (mobile) tractors, all openings into equipment mounted on instrument panels shall be sealed with tape conforming to PPP-T-60, type IV, class 1. In addition, for type II unboxed (mobile) tractors that are not equipped with closed cabs or the panel is not equipped with a vandal guard, the dial glasses of instruments mounted on the panels shall be individually covered or the entire panel may be covered when practicable, with a fitted piece of hardboard or 1/4-inch plywood secured in place with tape as specified herein.

3.4.1.19 Trouble lights, lamps, reflectors, and rear-vision mirrors. Trouble lights, removed lamps, removed reflectors, and removed rear-vision mirrors shall be individually placed in boxes conforming to PPP-B-636, W5c or W6c, and the boxes sealed in accordance with the appendix to the box specification, method V. Lamps, reflectors, and mirrors not removed from type II unboxed (mobile) tractors shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The barrier material shall be secured in place with tape conforming to PPP-T-60, type IV, class 1.

3.4.1.20 Gearshifts and control levers. Unpainted metal surfaces, handles and levers shall be coated with type P-1 preservative. For type II unboxed (mobile) tractors, composition and rubber handgrips (if not enclosed within a full cab)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

shall be removed and preserved in accordance with MIL-P-116, method IC-1, and placed in the tractor toolbox, or shall be consolidated as specified in 3.4.1.33. The threaded surfaces, exposed by removal of handgrips, shall be coated with type P-1 preservative. When the composition or rubber handgrips are left in place for any reason the grips shall be wrapped with the barrier material conforming to PPP-B-1055, class E-2 and the wrap secured in place with tape conforming to PPP-T-60, type IV, class 1.

3.4.1.21 Drivebelts and pulleys. Drivebelts shall be removed or released from tension. Unpainted pulley groves shall be coated with primer conforming to TT-P-664. Belts removed shall be preserved in accordance with MIL-P-116, method IC-1. A tag shall be attached in a conspicuous location within the operator's compartment indicating: "Drivebelts have been released from tension or removed. Check installation prior to operation." Tags shall be as specified in 3.4.1.3.

3.4.1.22 Seatbacks, armrests and cushions. Type II unboxed (mobile) tractors not equipped with closed cabs shall have the seatbacks, armrests and cushions that are removable, removed and individually wrapped with black polyethylene conforming to L-P-378 (6 mil). The plastic film shall be secured with tape conforming to PPP-T-60, type IV, class 1. The removed items shall be consolidated as specified in 3.4.1.33.2. Seatbacks, armrests and seat cushions that are not removable for type II unboxed (mobile) tractors not equipped with fully closed cabs, shall be wrapped or covered in place with the plastic film specified herein and the wrap secured with tape as specified herein. A 2-inch opening for drainage shall be made in the plastic film at the underside of seatbacks, armrests and cushions.

3.4.1.23 Windshield wiper arms and blades. For type II unboxed (mobile) tractors, windshield wiper arms and blades shall be removed and preserved in accordance with MIL-P-116, method IC-1. The preserved wiper arms and blades shall be placed in the toolbox or shall be consolidated as specified in 3.4.1.33. The exposed wiper shafts shall be coated with type P-1 preservative.

3.4.1.24 Locks, keys and key openings. Interiors of locks shall be coated with lubricant conforming to VV-L-800. Openings into exterior locks for type II unboxed (mobile) tractors shall be sealed with tape conforming to PPP-T-60, type IV, class 1. Switch keys, cab door keys, and padlocks with keys shall be preserved in accordance with MIL-P-116, method IC-1. The preserved keys and padlocks with keys shall be placed in the toolbox. A tag shall be attached in a conspicuous location in the operator's compartment indicating: "Keys for this tractor are preserved and placed in the toolbox." Tags shall be as specified in 3.4.1.3.

3.4.1.25 Hinges and fasteners. Hinges and metal fasteners shall be coated with type P-10, type I, grade 10 preservative.

3.4.1.26 Engine hoods. Engine hoods shall be closed and secured. For type II unboxed (mobile) tractors, vents, louvers, and other openings into the engine compartment shall be sealed with tape conforming to PPP-T-60, type IV, class 1, or the openings shall be covered with barrier material conforming to PPP-B-1055, class E-2, and the cover secured in place with tape as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

3.4.1.27 Radiator fronts. When specified (see 6.2), for type II unboxed (mobile) tractors, the radiator fronts shall be covered with barrier material conforming to PPP-B-1055, class E-2, and the barrier material secured in place with tape conforming to PPP-T-60, type IV, class 1.

3.4.1.28 Attachments.

3.4.1.28.1 Power takeoff and power control units. Power takeoff and power control units shall have the components preserved as specified herein for parts and assemblies of similar design and having similar characteristics.

3.4.1.28.2 Winches. Components of winches shall be preserved as specified herein for components of similar design and having similar characteristics.

3.4.1.28.3 Wire rope. Wire rope (except galvanized) shall be coated with type P-1 preservative applied by drawing the wire rope through a trough filled with preservative over an arrangement of three pulleys or by any other convenient manner so that the rope will be completely immersed in the preservative. Excess preservative shall be removed by a swab around a cable guide in the end of the trough. The wire rope shall be wound on the winch drums. For type II unboxed (mobile) tractors, the wire rope shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The barrier material shall extend around the rope on the drum and be of sufficient width to be taped to the flanges of the drum with tape conforming to PPP-T-60, Type IV, Class 1. A 2-inch opening for drainage shall be made in the barrier material at the underside of the drum.

3.4.1.28.4 Scarifiers. Scarifiers and components shall be preserved as specified herein for parts and assemblies having similar characteristics. Scarifiers mounted on tractors shall be positioned and secured in the transport position.

3.4.1.28.5 Other attachments. Components of other attachments such as dozers, rippers, loaders, pipe layers, angle fillers, mowers, sweepers, backhoes, cranes, and any other attachments shall be preserved as specified herein for components having similar characteristics.

3.4.1.29 Fire extinguishers. Unpainted exterior surfaces of fire extinguishers requiring the application of a contract preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. For type I boxed tractors, the fire extinguishers shall be secured in the mounting brackets. For Type II unboxed (mobile) tractors, each fire extinguisher shall be placed in a box conforming to PPP-B-636, W5c, or W6c. The boxes shall be waterproofed sealed in accordance with the appendix to the box specification, method V.

3.4.1.30 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A.

3.4.1.31 Technical publications. Technical publications for each tractor shall be preserved together in accordance with MIL-P-116, method IC-1.

3.4.1.32 Repair parts. Unless otherwise specified herein, the preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts. When specified, repair parts shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

preserved in accordance with MIL-S-196, level A; however, when parts are not specifically covered in MIL-S-196, an applicable submethod of preservation of MIL-P-116 shall be used.

3.4.1.33 Consolidation.

3.4.1.33.1 For Type I boxed tractors. The repair parts, tools, technical publications, and all other loose components (except wheels and attachments) requiring the protection of a container shall be consolidated. The toolbox shall be utilized as a container and the lid shall be closed and secured. The remainder of the components that will not fit in the toolbox shall be consolidated in close-fitting boxes conforming to PPP-B-601, domestic type, style optional, grade B, or wood boxes conforming to PPP-B-621, class 1, style optional. The number, size and weight of the boxes shall be determined by the available space for packing in the crate with the tractor in a manner to conserve cube.

3.4.1.33.2 For Type II unboxed (mobile) tractors. Consolidation shall be as specified in 3.4.1.33.1, except the boxes shall be overseas type, and the lid of the toolbox shall be secured to prevent pilferage. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as required. To the maximum extent possible, number, size and weight of the boxes shall be determined by the available space and convenience for securing to the tractor in a manner that will not increase cubage or interfere with towing or lifting the tractor with slings. Otherwise, the boxes shall be shipped separately from but at the same time as the tractor.

3.4.1.33.3 For attachments. Components removed from attachments shall be consolidated as specified in 3.4.1.33.1 for boxed attachments and as specified in 3.4.1.33.2 for unboxed attachments.

3.4.2 Level C. The tractor and all of the components shall be preserved as specified for level A with the following exceptions:

3.4.2.1 Matchmarking. Matchmarking shall be as specified in 3.4.1.2 for level A, except tags shall be as specified in 3.4.2.2.

3.4.2.2 Tags. Tags for matchmarking and instructions shall be as specified in 3.4.1.3 for level A, except marking ink need not be waterproof.

3.4.2.3 Enclosed gears and chains. The gear housing and chain housings shall be filled to the operating level with the applicable type and grade of lubricant required for operation. A tag as specified in 3.4.2.2 shall be attached to each housing in a conspicuous location indicating: "This housing is filled with the applicable type and grade of lubricant required for operation."

3.4.2.4 Exposed drive chains. Exposed drive chains shall be coated with type P-3 or P-9 preservative.

3.4.2.5 Clutches.

3.4.2.5.1 Disc-type. Spring-loaded-type clutches shall have the clutch pedal secured in a partially disengaged position. Snap-over-center- and toggle-in-type clutches shall be completely disengaged.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

3.4.2.5.2 Oil-type and hydraulic boosters. Oil-type flywheel and steering clutches and hydraulic boosters shall be preserved as specified for level A in 3.4.1.11.5, 3.4.1.11.6, 3.4.1.11.7.2, and 3.4.1.11.7.3, except tags shall be as specified in 3.4.2.2.

3.4.2.6 Hydraulic systems (including hydraulic brakes). Hydraulic systems (including hydraulic brakes) shall be preserved and tagged as specified in 3.4.1.12.2, 3.4.1.12.4 and 3.4.1.13, except tags shall be as specified in 3.4.2.2.

3.4.2.7 Airbrake systems. Airbrake systems shall be preserved and tagged as specified in 3.4.1.12.3 and 3.4.1.12.4, except tags shall be as specified in 3.4.2.2.

3.4.2.8 Automatic transmissions, fluid couplings and torque converter housings. Automatic transmissions, fluid couplings and torque converter housings, (except diesel-fuel operated torque converters) shall be filled to the operating level with the applicable type and grade of fluid or lubricant required for operation. A tag as specified in 3.4.2.2 shall be attached to each control indicating: "The housing is filled with the applicable type and grade of fluid or lubricant required for operation." Diesel-fuel-operated torque converters shall be preserved and tagged as specified in 3.4.1.14.2.2 for level A, except tags shall be as specified herein.

3.4.2.9 Drivebelts. Drivebelts shall be released from tension. A tag shall be attached in a conspicuous location within the operator's compartment indicating: "Tension has been released on all drivebelts. Adjust tension prior to operation." Tags shall be as specified in 3.4.2.2.

3.4.2.10 Windshield wiper arms and blades. For type II unboxed (mobile) tractors, windshield wiper arms and blades shall be removed and placed in the contractor's commercial box.

3.4.2.11 Locks and keys. Switch keys, cab door keys and padlocks with keys shall be placed in the toolbox. A tag shall be attached in a conspicuous location in the operator's compartment indicating: "Keys for this tractor are placed in the toolbox." Tags shall be as specified in 3.4.2.2.

3.4.2.12 Fire extinguishers. Fire extinguishers shall be secured in their mounting brackets.

3.4.2.13 Maintenance tools. Maintenance tools, shall be preserved as specified in PPP-P-40, level C requirements.

3.4.2.14 Other components. Those components that do not require preservation as specified for level A include the following: the drive shafts and universal joints, clutch housings, wheels, axles and bearings, crawlers, gear shifts and control levers and the attachments. These components shall be preserved in a manner to prevent deterioration and damage during handling and shipment.

3.4.2.15 Consolidation. Consolidation of components removed from tractors, and attachments shall be as specified in 3.4.1.33.1 and 3.4.1.33.2 for level A, except in all cases, boxes shall be domestic type and strapping of boxes for mobile shipment may be finish A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

3.5 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.5.1 Level A.

3.5.1.1 Type I boxed tractors. The tractor, and attachments that are practical to pack with the tractor, consolidated items and all other components, comprising each complete tractor shall be packed together in a crate conforming to MIL-C-104, class I, style A or B, as applicable. The tractor and components, including wheels if removed, attachments, and consolidated containers shall be securely blocked, braced and anchored within the crate in accordance with MIL-STD-1186. Attachments such as dozers, buckets, booms, or cranes, which are impractical to pack in the crate with the tractor, shall be packed for shipment as separate items as specified in 3.5.1.3. The attachments packed as separate items shall be shipped at the same time as the tractor.

3.5.1.2 Type II unboxed (mobile) tractors. The tractor shall be prepared for unboxed shipment. Consolidated containers and other components shall be positioned on the tractor in any available space that will not increase cubage, damage the tractor, or interfere with towing, lifting with slings, or steering the tractor while being towed. Containers and components shall be secured to the tractor with strapping conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable, or secured by bolting. When specified (see 6.2), attachments shall be positioned in their normal operating or transport position, whichever is the most convenient for handling and shipment, and shall be secured in position with flat steel strapping as specified herein, or with wire rope. In addition, the attachments shall be blocked and braced as required to prevent movement or damage. Attachments such as dozers, buckets and booms that cannot be relocated on and secured to the tractor shall be packed as specified in 3.5.1.3. The attachments packed as separate items shall be shipped at the same time as the tractor.

3.5.1.2.1 Cabs. Cab windows and doors shall be closed. When specified (see 6.2), windshields and window glass shall be protected from breakage by covering each glass surface with a fitted piece of hardboard, or plywood 1/4-inch thick, or with a sheet metal panel secured in place. There shall be not less than a 1-inch clearance between the panels and the glass. Small openings into the cabs shall be sealed with tape conforming to PPP-T-60, type IV, class 1.

3.5.1.3 Attachments. Unless otherwise specified, attachments such as dozers, buckets and booms shall be prepared for shipment unboxed. To the maximum extent possible, related parts and consolidated components shall be secured to the attachments with steel strapping as specified herein. Boxed and unboxed attachments and consolidated items that cannot be secured to the attachment or tractor shall be shipped separately from, but at the same time as the tractor. When specified (see 6.2), attachments such as dozers, treedozers, rippers, loaders, pipelayers, mowers, sweepers, cranes, component parts, and consolidated items comprising complete attachments shall be packed in crates conforming to MIL-C-3774, type I. The contents of the crates shall be blocked, braced and anchored in accordance with MIL-STD-1186. Strapping used for securing of the crate contents shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

3.5.2 Level B.

3.5.2.1 Type I boxed tractors. The tractor and its components shall be packed as specified in 3.5.1.1 for level A. As an alternate, crates shall conform to MIL-C-3774, type I.

3.5.2.2 Type II unboxed (mobile) tractors. The tractor and components shall be packed as specified in 3.5.1.2. Strapping may be finish A.

3.5.3 Level C. The tractors, components, attachments, repair parts, tools, and technical publications shall be prepared for unboxed (mobile) shipment to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

3.6 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

3.7 Depreservation guides. Two depreservation guides shall be prepared. One guide shall be placed in a waterproof envelope marked "Depreservation Guide" and secured in a conspicuous location on the steering wheel or steering controls of the tractor. The other guide shall be placed in the package of technical manuals and the package marked to indicated the depreservation guide is inside. Unless otherwise specified (see 6.2), DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used.

3.8 Workmanship. All operations and processes involved in accomplishing the cleaning, preserving, packaging, packing, and marking requirements specified herein shall be in accordance with the highest grade practices associated with this type of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article pack inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article pack inspection.

4.3.1 Examination. The first article pack shall be examined for the defects specified in 4.4.4. Examination shall be in three stages specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article pack of boxed tractors and of boxed attachments for level A or B shall be tested as follows: The boxed tractor and the boxed attachments shall be subjected to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles per hour. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visible damage to the tractors or attachments shall constitute failure of the first article pack and shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection stages.

a. The first stage shall include inspection of procedures, containers, materials, methods, marking, and placement of components in consolidated containers.
b. The second stage shall include inspection of the marking and closure of the consolidated containers and the securing of the containers and components to the shipping crates and unboxed (mobile) tractors.
c. The third stage shall include inspection of materials and marking of uncrated items and crated items after closure.

4.4.2 Unit of product. For the purpose of inspection, a completed pack prepared for any applicable inspection stage shall be considered a unit of product.

4.4.3 Sampling. Sample size shall be determined by using MIL-STD-105, table I and IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of Defects

Defect	Level A B C	Requirement Paragraph
101. Materials, methods and containers not as specified. Each incorrect material, method or container shall constitute one defect.	X X X	3.1, 3.4, 3.5, 3.6
102. Damaged portions of painted surfaces not repainted as specified.	X - -	3.3
103. Disassembly not as specified.	X - X	3.4.1.1, 3.4.2
104. Matchmarking not as specified.	X - X	3.4.1.2, 3.4.2.1
105. Tags used for matchmarking not as specified.	X - X	3.4.1.3, 3.4.2.2
106. Printing on tags not waterproof.	X - -	3.4.1.3
107. Unprotected metal surfaces not coated with preservatives as specified.	X - X	3.4.1.5, 3.4.2
108. Preserved machined surfaces not wrapped or covered with barrier material, and the wraps or covers not secured with tape as specified.	X - X	3.4.1.5, 3.4.2
109. Engines and engine components not preserved as specified.	X - X	3.4.1.6, 3.4.2
110. Pumps, burners and torches not preserved as specified.	X - X	3.4.1.7.1, 3.4.2
111. Blowtorches and blowtorch type burners not placed in fiberboard boxes as specified.	X - X	3.4.1.7.1, 3.4.2
112. Heater fuel tanks not preserved as specified.	X - X	3.4.1.7.2, 3.4.2
113. Electric motors not preserved in accordance with referenced document.	X - X	3.4.1.7.3, 3.4.2
114. Drive shafts and universal joints not coated with preservative as specified and dust boots not reinstalled as specified.	X - -	3.4.1.8
115. Exposed gears not coated with preservative or with primer as specified.	X - X	3.4.1.9.1, 3.4.2
116. Enclosed gear housing not filled with preservative oils or with operational lubricant as specified and housing not tagged as specified.	X - X	3.4.1.9.2, 3.4.2.3
117. Exposed drive chains not coated with preservative as specified.	X - X	3.4.1.10.1, 3.4.2.4
118. Enclosed chain housing not filled with preservative oil or not filled with operational lubricant as applicable, and the housing not tagged as specified.	X - X	3.4.1.10.2, 3.4.2.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of Defects. (continued)

Defect	Level A B C	Requirement Paragraph
119. Dry-type clutch components not coated as specified.	X - -	3.4.1.11.1, 3.4.1.11.2, 3.4.1.11.3, 3.4.1.11.4
120. Disc-type and cone-type clutches not partially or fully disengaged as specified.	X - X	3.4.1.11.1, 3.4.1.11.3, 3.4.2.5.1
121. Flywheel clutches not preserved and tagged as specified.	X - X	3.4.1.11.5, 3.4.1.11.6, 3.4.2.5.2
122. Steering clutches not preserved and not tagged as specified.	X - X	3.4.1.11.7, 3.4.2.5.2
123. Hydraulic boosters not preserved and tagged as specified.	X - X	3.4.1.11.7.3, 3.4.2.5.2
124. Clutch housing drainplugs for dry-type clutches for type I boxed tractors not removed, not preserved, and not tagged as specified.	X - -	3.4.1.11.8
125. Openings into clutch housings for type II unboxed (mobile) tractors not covered and not tagged as specified.	X - -	3.4.1.11.8
126. Exterior surfaces of brakes not coated with preservative as specified.	X - X	3.4.1.12.1, 3.4.2
127. Surfaces of brake components enclosed within brakedrums and the braking surfaces of brakedrums not coated with primer when specified and not tagged as specified.	X - X	3.4.1.12.1, 3.4.2
128. Hydraulic brake system operating on nonpetroleum base fluid not filled with preservative fluid not tagged as specified.	X - X	3.4.1.12.2.1, 3.4.2.5.2
129. Hydraulic brake systems operating on petroleum base fluid not filled to the operating level with the fluid required for operation and the system not tagged as specified.	X - X	3.4.1.12.2.2, 3.4.2.5.2
130. Air-actuated brake system not preserved and not tagged as specified. ified.	X - X	3.4.1.12.3, 3.4.2.7
131. Air hydraulic brakes not preserved and not tagged as specified.	X - X	3.4.1.12.4, 3.4.2.7
132. Automatic transmissions, fluid couplings, and torque converters not preserved and tagged as specified or not filled with	X - X	3.4.1.14, 3.4.2.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of Defects. (continued)

Defect	Level A B C	Requirement Paragraph
applicable type lubricant and tagged as specified.		
133. Wheels, axles and bearings not preserved as specified.	X - -	3.4.1.15
134. Rubber tires not inflated to pressure specified.	X - X	3.4.1.16, 3.4.2
135. Crawler components not coated with preservative as specified.	X - -	3.4.1.17
136. Instrument panels for type II unboxed (mobile) tractors not protected as specified.	X - X	3.4.1.18, 3.4.2
137. Trouble lights, lamps, reflectors and rear-vision mirrors not boxed and, for type II unboxed (mobile) tractors, not protected as specified.	X - X	3.4.1.19, 3.4.2
138. Fiberboard boxes not closed as specified.	X - X	3.4.1.7.1, 3.4.1.11, 3.4.1.29, 3.4.2
139. Gearshifts and control levers not preserved as specified and, for type II unboxed (mobile) tractors, not equipped with a full cab, composition and rubber handgrips not preserved or protected as specified.	X - -	3.4.1.20
140. Drivebelts not removed or released from tension and not tagged as specified.	X - X	3.4.1.21 3.4.2.9
141. Pulley grooves not coated with primer as specified.	X - -	3.4.1.21
142. Seatbacks, armrests and cushions for type II unboxed (mobile) tractors not equipped with closed cabs not wrapped or protected as specified.	X - X	3.4.1.22, 3.4.2
143. Windshield wiper arms and blades for type II unboxed (mobile) tractors not removed and preserved as specified and wiper shafts not coated with preservative as specified	X - X	3.4.1.23, 3.4.2.10
144. Interior of locks not coated with lubricant and keys not preserved as specified.	X - -	3.4.1.24
145. Openings into exterior locks for type II unboxed (mobile) tractors not sealed with tape as specified.	X - -	3.4.1.24

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of Defects. (continued)

Defect	Level A B C	Requirement Paragraph
146. Keys and padlocks not in toolbox as specified.	X - X	3.4.1.24, 3.4.2.11
147. Tag not attached in operator's compartment indicating keys and padlocks are in toolbox.	X - X	3.4.1.24 3.4.2.11
148. Hinges and fasteners not coated with preservative as specified.	X - X	3.4.1.25, 3.4.2
149. Engine hoods not closed and secured as specified.	X - X	3.4.1.26, 3.4.2
150. Vents, louvers and other openings into engine hoods not covered for type II unboxed (mobile) tractors.	X - X	3.4.1.27, 3.4.2
151. Radiator fronts for type II unboxed (mobile) tractors not covered when specified.	X - -	3.4.1.27
152. Attachments not preserved, and positioned as specified.	X - -	3.4.1.28.1 3.4.1.28.2, 3.4.1.28.4, 3.4.1.28.5
153. Wire rope (except galvanized) not coated with preservative as specified and not rewound on drum as specified.	X - -	3.4.1.28.3
154. Wire rope for type II unboxed (mobile) tractors not covered as specified.	X - -	3.4.1.28.3
extinguishers not coated with preservative as specified.		
156. Fire extinguishers not secured in mounting brackets or placed in a box as specified.	X - X	3.4.1.29, 3.4.2.12
157. Maintenance tools not preserved in accordance with referenced document.	X - X	3.4.1.30, 3.4.2.13
158. Technical publications not preserved as specified.	X - X	3.4.1.31, 3.4.2.13
159. Repair parts not preserved in accordance with the referenced document.	X - X	3.4.1.32, 3.4.2.13
160. Consolidation not as specified.	X - X	3.4.1.33, 3.4.2.15
161. Toolbox lids not secured to prevent pilferage.	X - X	3.4.1.33, 3.4.2.15
162. Packing not as specified.	X X X	3.5
163. Blocking, bracing and anchoring not as specified.	X X -	3.5.1.1, 3.5.1.3, 3.5.2.1
164. Positioning of consolidated components on tractors prepared	X X -	3.5.1.2, 3.5.2.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of Defects. (continued)

Defect	Level A B C	Requirement Paragraph
for unboxed (mobile) shipment not as specified.		
165. Attachments not positioned as specified, when required, for type II, unboxed (mobile) tractors.	X X -	3.5.1.2, 3.5.2.2
166. Windshields and window glass not protected as specified, when required, and cab openings not sealed as specified.	X X -	3.5.1.2.1, 3.5.2.2
167. Strapping not as specified.	X X -	3.4.1.33.2, 3.4.1.33.3, 3.4.2.15, 3.5.1.2, 3.5.1.3, 3.5.2.2
168. Marking missing, illegible, incomplete, or incorrect.	X X X	3.6
169. Depreservation guides not prepared and stowed as specified.	X X -	3.7

5. PACKAGING

(This section not applicable to this specification).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. It is intended that this specification cover preservation, packing, and marking of tractors and their attachments, and shall be used for reference in section 5 of equipment specifications, direct references in contract or orders, or in the preparation of packaging data sheets.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type of shipment required (see 1.2).
d. Time frame required for submission of the first article pack (see 3.2).
e. Preservation and packing required (see 3.4 and 3.5).
f. When disassembly of the clutch is required for preservation (see 3.4.1.11.1).
g. When brake components enclosed within brakedrums and the braking surfaces of brakedrums are to be preserved (see. 3.4.1.12.1).
h. When radiator fronts are to be protected (see 3.4.1.27).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

i. When repair parts should be preserved in accordance with MIL-S-196 (see 3.4.1.27).
j. When attachments are to be positioned in normal or transport position (see 3.5.1.2).
k. When windshields and window glass are to be protected (see 3.5.1.2.1).
l. When attachments are to be prepared for boxed shipment (see 3.5.1.3).
m. Any special marking required (see 3.6).
n. When other than DA Form 2258 is to be used (see 3.7).

6.3 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the supplier of his obligation to preserve, pack, and mark the tractors and attachments in accordance with this specification.

6.4 Painting of the end item. The requirements of 3.3 are intended to apply only in those cases where the painting requirements of the equipment specification would not otherwise be applicable, such as on separate packaging contracts or in deport operations.

6.5 Percentage of fuel oil. The percentage of fuel oil contained in preservative oil which has been used to preserve burner fuel tanks (see 3.4.1.7.2) may be determined within practical limits by use of American Petroleum Institute hydrometers and control specimens of known dilution.

6.6 Depreservation guide. The contracting officer should arrange to furnish the necessary copies of DA Form 2258 when required by the supplier (see 3.7).

6.7 Environmental pollution preventive measures. Environmental pollution preventive measures are contained in the material specifications indicated herein. Refer to material specifications (or preparing activity) for recommended disposability methods.

6.8 Subject term (key word) listing.

Marking
Preservation

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

Custodians:

Army - ME
Navy - YD
Air Force - 84

Review activities:

Army - SM, AT
Air Force - 99

User activity:

Navy - MC

Preparing activity:

Army - ME

(Project PACK-0918)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-3351E	900222

3. DOCUMENT TITLE: Tractor, Full-Tracked, Low-Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial; and Their Attachments, Packaging of

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:32Z
Check the source to verify that this is the current version before use.